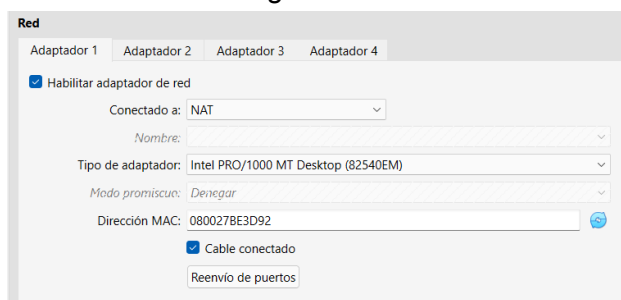


Guia pràctica: Servidor FTP i Servidor sFTP

0. Configuració de la màquina virtual i xarxa

Abans de començar amb els serveis, es comprova que la màquina virtual té connectivitat correcta.

- Mode de xarxa configurat en NAT.



- Es verifica que la màquina respon correctament amb ping 8.8.8.8

```
usuari@server:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=31.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=30.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=31.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=30.6 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=30.7 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=36.5 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=30.7 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6006ms
rtt min/avg/max/mdev = 30.614/31.655/36.508/1.995 ms
usuari@server:~$
```

Activitat A – Servidor FTP (vsftpd)

1. Instal·lació del servidor FTP

Instal·lació del servidor FTP vsftpd:

sudo apt update

sudo apt install vsftpd

```
usuari@server:~$ sudo apt install vsftpd
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
vsftpd ya está en su versión más reciente (3.0.5-0ubuntu3.1).
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 61 no actualizados.
usuari@server:~$
```

2. Configuració de l'accés anònim

Per defecte, el servidor FTP no permet connexions anònimes.

Editem el fitxer de configuració:

sudo nano /etc/vsftpd.conf

Opcions utilitzades:

anonymous_enable=YES

anon_root=/srv/ftp/public

```
GNU nano 7.2 /etc/vsftpd.conf *
# daemon started from an initscript.
listen=NO
#
# This directive enables listening on IPv6 sockets. By default, listening
# on the IPv6 "any" address (:::) will accept connections from both IPv6
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=YES
anon_root=/srv/ftp/public
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
```

Es reinicia el servei:

sudo systemctl restart vsftpd

Proves en mode anònim

- Creem una carpeta i un arxiu de prova
- Connexió com a anonymous

```
usuari@server:~$ ftp 127.0.0.1
Connected to 127.0.0.1.
220 (vsFTPD 3.0.5)
Name (127.0.0.1:usuari): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

- Descàrrega de fitxers correcta

```
ftp> get prova.txt
local: prova.txt remote: prova.txt
229 Entering Extended Passive Mode (||||11546|)
150 Opening BINARY mode data connection for prova.txt (20 bytes).
100% |*****|
226 Transfer complete.
20 bytes received in 00:00 (33.96 KiB/s)
ftp>
```

- Intent de pujada amb error de permisos

```
ftp> put pujada.txt
local: pujada.txt remote: pujada.txt
229 Entering Extended Passive Mode (||||25898|)
550 Permission denied.
ftp>
```

3. Usuaris autenticats i carpeta de treball

Creació de l'usuari local:

`sudo adduser albert`

```
usuari@server:~$ sudo adduser albert
[sudo] password for usuari:
info: Adding user `albert' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `albert' (1001) ...
info: Adding new user `albert' (1001) with group `albert (1001)' ...
info: Creating home directory `/home/albert' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for albert
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `albert' to supplemental / extra groups `users' ...
info: Adding user `albert' to group `users' ...
usuari@server:~$ _
```

Creació de la carpeta de treball:

`sudo mkdir /home/albert/ftp`

`sudo chown albert:albert /home/albert/ftp`

```
usuari@server:~$ sudo mkdir /home/albert/ftp
usuari@server:~$ sudo chown albert:albert /home/albert/ftp
```

4. Engabiament (chroot) i permisos

Configuració de l'engabiament d'usuaris locals.

Editem el fitxer:

`sudo nano /etc/vsftpd.conf`

Opcions afegides:

`local_enable=YES`

`write_enable=YES`

`chroot_local_user=YES`

`allow_writeable_chroot=YES`

```
GNU nano 7.2 /etc/vsftpd.conf *
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=YES
anon_root=/srv/ftp/public
#
# Uncomment this to allow local users to log in.
local_enable=YES
write_enable=YES
chroot_local_user=YES
allow_writeable_chroot=YES
#
# Uncomment this to enable any form of FTP write command
```

5. Reinici i prova de connexió FTP

`sudo systemctl restart vsftpd`

Connexió amb usuari autènticat:

`ftp 127.0.0.1`

```
usuari@server:~$ ftp 127.0.0.1
Connected to 127.0.0.1.
220 (vsFTPd 3.0.5)
Name (127.0.0.1:usuari): albert
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files
ftp>
```

Es comprova que l'usuari pot treballar només dins la seva carpeta.

```
ftp> pwd
Remote directory: /
ftp> put /etc/hosts
local: /etc/hosts remote: /etc/hosts
229 Entering Extended Passive Mode (|||49194|)
553 Could not create file.
ftp> mkdir test
257 "/test" created
```

6. Comprovació de dades en clar amb Wireshark (desde el mateix ubuntu)

Instal·lem tshark

`sudo apt install tshark`

```
usuari@server:~$ sudo apt install tshark
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
tshark ya está en su versión más reciente (4.2.2-1.1build3).
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 61 no actualizados.
usuari@server:~$
```

Comencem a capturar paquets

Per començar a capturar només FTP:

```
sudo tshark -i lo -f "tcp port 21" -w /tmp/captura_ftp.pcap
```

-f "tcp port 21" , captura només FTP

-w captura_ftp.pcap , guarda la captura en un fitxer per analitzar després

```
usuari@server:~$ sudo tshark -i lo -f "tcp port 21" -w /tmp/captura_ftp.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
```

Ens connectem al servidor FTP des de la mateixa VM

Obrim un altre terminal (Ctrl+Alt+F2) i ens connectem amb l'usuari albert:

Analitza els paquets

```
sudo tshark -r /tmp/captura_ftp.pcap -Y ftp
```

Això mostrarà només els paquets FTP.

```
usuari@server:~$ sudo tshark -r /tmp/captura_ftp.pcap -Y ftp
Running as user "root" and group "root". This could be dangerous.
  4 0.001310424    127.0.0.1 → 127.0.0.1    FTP 86 Response: 220 (vsFTPd 3.0.5)
  6 2.238866357    127.0.0.1 → 127.0.0.1    FTP 79 Request: USER albert
  8 2.239068856    127.0.0.1 → 127.0.0.1    FTP 100 Response: 331 Please specify the password.
 10 4.323407109    127.0.0.1 → 127.0.0.1    FTP 74 Request: PASS a
 11 4.347484562    127.0.0.1 → 127.0.0.1    FTP 89 Response: 230 Login successful.
 13 4.347541028    127.0.0.1 → 127.0.0.1    FTP 72 Request: SYST
 14 4.347654657    127.0.0.1 → 127.0.0.1    FTP 85 Response: 215 UNIX Type: L8
 15 4.347948249    127.0.0.1 → 127.0.0.1    FTP 72 Request: FEAT
 16 4.347963353    127.0.0.1 → 127.0.0.1    FTP 81 Response: 211-Features:
 17 4.347972290    127.0.0.1 → 127.0.0.1    FTP 73 Response:  EPRT
 19 4.347981172    127.0.0.1 → 127.0.0.1    FTP 73 Response:  EPSV
 20 4.347987221    127.0.0.1 → 127.0.0.1    FTP 73 Response:  MDTM
 22 4.347994258    127.0.0.1 → 127.0.0.1    FTP 73 Response:  PASV
 23 4.347999990    127.0.0.1 → 127.0.0.1    FTP 80 Response:  REST STREAM
 25 4.348007926    127.0.0.1 → 127.0.0.1    FTP 73 Response:  SIZE
 26 4.348013467    127.0.0.1 → 127.0.0.1    FTP 73 Response:  TVFS
 28 4.348020594    127.0.0.1 → 127.0.0.1    FTP 75 Response: 211 End
 30 22.967107127    127.0.0.1 → 127.0.0.1    FTP 72 Request: QUIT
 31 22.967155470    127.0.0.1 → 127.0.0.1    FTP 80 Response: 221 Goodbye.
usuari@server:~$
```

Podem veure en text pla:

Nom d'usuari

Contrasenya

Comandes

Conclusió implícita: FTP no és segur.

Activitat B – Servidor sFTP (via SSH)

1. Servei SSH

Es comprova que el servei SSH està actiu:

No cal instal·lar res nou. SSH ja ve instal·lat:

`sudo systemctl status ssh`

```
usuari@server:/$ sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: inactive (dead)
 TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
usuari@server:/$
```

2. Gestió d'usuaris i grups

Creació de grups:

`sudo addgroup sftpushers`

`sudo addgroup sysadmin`

Creació i assignació d'usuaris:

`sudo adduser albertssh`

`sudo usermod -aG sftpushers albertssh`

```
usuari@server:/$ sudo addgroup sftpushers
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `sftpushers' (GID 1002) ...
usuari@server:/$ sudo addgroup sysadmin
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `sysadmin' (GID 1003) ...
usuari@server:/$ sudo adduser albertssh
info: Adding user `albertssh' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `albertssh' (1004) ...
info: Adding new user `albertssh' (1004) with group `albertssh (1004)' ...
info: Creating home directory `/home/albertssh' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for albertssh
Enter the new value, or press ENTER for the default
  Full Name []:
   Room Number []:
    Work Phone []:
    Home Phone []:
      Other []:
Is the information correct? [Y/n] Y
info: Adding new user `albertssh' to supplemental / extra groups `users' ...
info: Adding user `albertssh' to group `users' ...
usuari@server:/$ sudo usermod -aG sftpushers albertssh
usuari@server:/$ _
```

3. Estructura de directoris

Creació de l'estructura necessària per a l'engabament:

```
sudo mkdir -p /sftp/albertssh/files
sudo chown root:root /sftp/albertssh
sudo chmod 755 /sftp/albertssh
sudo chown albertssh:albertssh /sftp/albertssh/files
```

```
usuari@server:/$ sudo mkdir -p /sftp/albertssh/files
usuari@server:/$ sudo chown root:root /sftp/albertssh
usuari@server:/$ sudo chmod 755 /sftp/albertssh
usuari@server:/$ sudo chown albertssh:albertssh /sftp/albertssh/files
usuari@server:/$ _
```

4. Configuració d'SSH per a sFTP

Editem el fitxer de configuració:

```
sudo nano /etc/ssh/sshd_config
```

Configuració aplicada per als usuaris limitats:

```
Match Group sftputers
  ChrootDirectory /sftp/%u
  ForceCommand internal-sftp
  AllowTCPForwarding no
  X11Forwarding no
```

```
GNU nano 7.2 /etc/ssh/sshd_config *
Match Group sftputers
  ChrootDirectory /sftp/%u
  ForceCommand internal-sftp
  AllowTCPForwarding no
  X11Forwarding no
# This is the sshd server system-wide configuration file. See
```

5. Reinici i proves finals

`sudo systemctl restart ssh`

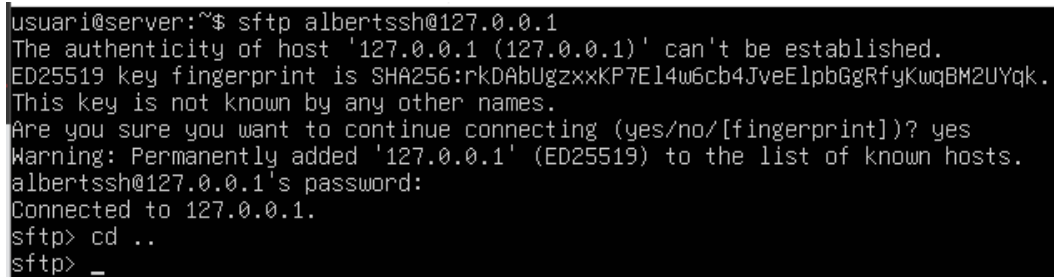
Prova d'usuari estàndard

Connexió amb l'usuari albertssh:

`sftp albertssh@127.0.0.1`

No es permet sortir de la carpeta assignada

`cd ..`

A terminal window showing the process of connecting to an sftp server. The user runs 'sftp albertssh@127.0.0.1'. The terminal displays a warning about the host's authenticity, showing a key fingerprint. The user responds 'yes' to continue. The terminal then shows the user's password being entered, the connection being established, and the user attempting to change directories with 'cd ..' and 'cd _', both of which fail as indicated by the prompt returning to the initial state.

```
usuari@server:~$ sftp albertssh@127.0.0.1
The authenticity of host '127.0.0.1 (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:rkDAbUgzxxKP7El4w6cb4JveElpbGgRfyKwqBM2UYqk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '127.0.0.1' (ED25519) to the list of known hosts.
albertssh@127.0.0.1's password:
Connected to 127.0.0.1.
sftp> cd ..
sftp> _
```

Comprovació de dades sFTP amb Wireshark

Captura paquets amb tshark:

`sudo tshark -i lo -f "tcp port 22" -w /tmp/captura_sftp.pcap`

Torna al terminal i fer la connexió sFTP i transferència:

`sftp albertssh@127.0.0.1`

Analitza amb:

`sudo tshark -r /tmp/captura_sftp.pcap -Y "tcp.port == 22"`


```
28
3 0.000021752 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2030677481 TSecr=2030677481
4 0.000118652 127.0.0.1 → 127.0.0.1 SSH 109 Client: Protocol (SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14)
5 0.000121222 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=1 Ack=44 Win=65536 Len=0 TSval=2030677481 TSecr=2030677481
6 0.005007757 127.0.0.1 → 127.0.0.1 SSHv2 109 Server: Protocol (SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14)
7 0.005015442 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=44 Ack=44 Win=65536 Len=0 TSval=2030677486 TSecr=2030677486
8 0.005759936 127.0.0.1 → 127.0.0.1 SSHv2 1186 Server: Key Exchange Init
9 0.005763863 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=44 Ack=1164 Win=69120 Len=0 TSval=2030677487 TSecr=2030677487
10 0.005801048 127.0.0.1 → 127.0.0.1 SSHv2 1602 Client: Key Exchange Init
11 0.046536931 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=1164 Ack=1580 Win=78976 Len=0 TSval=2030677528 TSecr=2030677487
12 0.046543173 127.0.0.1 → 127.0.0.1 SSHv2 1274 Client: Diffie-Hellman Key Exchange Init
13 0.046547652 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=1164 Ack=2788 Win=79104 Len=0 TSval=2030677528 TSecr=2030677528
14 0.053928209 127.0.0.1 → 127.0.0.1 SSHv2 1598 Server: Diffie-Hellman Key Exchange Reply, New Keys
15 0.071959603 127.0.0.1 → 127.0.0.1 SSHv2 150 Client: New Keys
16 0.112277745 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=2696 Ack=2872 Win=79104 Len=0 TSval=2030677594 TSecr=2030677553
17 0.112286235 127.0.0.1 → 127.0.0.1 SSHv2 110 Client:
18 0.112293853 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=2696 Ack=2916 Win=79104 Len=0 TSval=2030677594 TSecr=2030677594
19 0.112329863 127.0.0.1 → 127.0.0.1 SSHv2 110 Server:
20 0.112350224 127.0.0.1 → 127.0.0.1 SSHv2 134 Client:
21 0.112755757 127.0.0.1 → 127.0.0.1 SSHv2 330 Server:
22 0.153693995 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=2984 Ack=3004 Win=79744 Len=0 TSval=2030677635 TSecr=2030677594
23 2.036657449 127.0.0.1 → 127.0.0.1 SSHv2 150 Client:
24 2.072810167 127.0.0.1 → 127.0.0.1 SSHv2 94 Server:
25 2.072821039 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=3068 Ack=3032 Win=79744 Len=0 TSval=2030679554 TSecr=2030679554
26 2.072863656 127.0.0.1 → 127.0.0.1 SSHv2 178 Client:
27 2.113915757 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=3032 Ack=3180 Win=79104 Len=0 TSval=2030679595 TSecr=2030679554
28 2.182588622 127.0.0.1 → 127.0.0.1 SSHv2 694 Server:
29 2.223683671 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=3180 Ack=3660 Win=79744 Len=0 TSval=2030679705 TSecr=2030679664
30 2.223689760 127.0.0.1 → 127.0.0.1 SSHv2 110 Server:
31 2.223695078 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=3180 Ack=3704 Win=79744 Len=0 TSval=2030679705 TSecr=2030679705
32 2.223734658 127.0.0.1 → 127.0.0.1 SSHv2 186 Client:
33 2.223737387 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=3704 Ack=3300 Win=79104 Len=0 TSval=2030679705 TSecr=2030679705
34 2.223944409 127.0.0.1 → 127.0.0.1 SSHv2 138 Server:
35 2.223979380 127.0.0.1 → 127.0.0.1 SSHv2 110 Client:
36 2.225417872 127.0.0.1 → 127.0.0.1 SSHv2 422 Server:
37 2.225480879 127.0.0.1 → 127.0.0.1 SSHv2 134 Client:
38 2.225500681 127.0.0.1 → 127.0.0.1 SSHv2 142 Server:
39 2.226334677 127.0.0.1 → 127.0.0.1 SSHv2 118 Client:
40 2.226379256 127.0.0.1 → 127.0.0.1 SSHv2 158 Server:
41 2.267294344 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=3464 Ack=4300 Win=79744 Len=0 TSval=2030679749 TSecr=2030679708
42 9.053535909 127.0.0.1 → 127.0.0.1 SSHv2 102 Client:
43 9.053718800 127.0.0.1 → 127.0.0.1 SSHv2 190 Server:
44 9.053725445 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=3500 Ack=4424 Win=79744 Len=0 TSval=2030686535 TSecr=2030686535
45 9.053750286 127.0.0.1 → 127.0.0.1 SSHv2 102 Client:
46 9.053771209 127.0.0.1 → 127.0.0.1 SSHv2 126 Client:
47 9.053777679 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [FIN, ACK] Seq=3596 Ack=4424 Win=79744 Len=0 TSval=2030686535 TSecr=2030686535
48 9.054193805 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [ACK] Seq=4424 Ack=3597 Win=79104 Len=0 TSval=2030686535 TSecr=2030686535
49 9.057722485 127.0.0.1 → 127.0.0.1 TCP 66 22 → 42744 [FIN, ACK] Seq=4424 Ack=3597 Win=79104 Len=0 TSval=2030686539 TSecr=2030686535
50 9.057733103 127.0.0.1 → 127.0.0.1 TCP 66 42744 → 22 [ACK] Seq=3597 Ack=4425 Win=79744 Len=0 TSval=2030686539 TSecr=2030686539
jsuar1@server:~$
```

```
sudo tshark -r /tmp/captura_sftp.pcap -x -Y "tcp.port == 22"
```

```
00b0  d3 b0 71 4d 1a df 4a 2b c8 4a d0 38 b4 27  ..qM..J+.J.8.'
0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 08  .....E.
0010  00 34 13 31 40 00 40 06 29 89 7f 00 00 01 7f 00  ..4.1@.@.).....
0020  00 01 a6 f8 00 16 95 a5 f0 e1 30 dc ab 71 80 10  .....0...q..
0030  02 6f fe 28 00 00 01 01 08 0a 79 09 d1 47 79 09  ..o.(.....y..Gy.
0040  d1 47  ..G
0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 08  .....E.
0010  00 58 13 32 40 00 40 06 29 64 7f 00 00 01 7f 00  ..X.2@.@.)d.....
0020  00 01 a6 f8 00 16 95 a5 f0 e1 30 dc ab 71 80 18  .....0...q..
0030  02 6f fe 4c 00 00 01 01 08 0a 79 09 d1 47 79 09  ..o.L.....y..Gy.
0040  d1 47 91 dd ec 49 e2 58 32 4c dc 28 4f c8 8e 65  ..G...I.X2L.(0..e
0050  5b d8 86 60 f4 08 40 1a f8 ab 87 52 42 3f de ba  [...`..@....RB?...
0060  4e ec 95 eb 33 5b  N...3[
0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 08  .....E.
0010  00 70 13 33 40 00 40 06 29 4b 7f 00 00 01 7f 00  ..p.3@.@.)K.....
0020  00 01 a6 f8 00 16 95 a5 f1 05 30 dc ab 71 80 18  .....0...q..
0030  02 6f fe 64 00 00 01 01 08 0a 79 09 d1 47 79 09  ..o.d.....y..Gy.
0040  d1 47 d2 c2 fd 2f cb 30 09 e1 f2 57 57 ee c0 fc  ..G.../.0...WW...
0050  c2 66 36 07 63 86 49 12 93 c8 04 7b 6a fd 04 f5  ..f6.c.I....{j...
0060  d9 0e a3 0a 0b 2b 39 9e dc 95 4c 63 c6 0a ee 03  .....+9...Lc....
0070  b3 8d bf f2 ec f0 ee 49 3a 3f dd 77 b5 b7  .....I:?.w..
0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 08  .....E.
0010  00 34 13 34 40 00 40 06 29 86 7f 00 00 01 7f 00  ..4.4@.@.).....
0020  00 01 a6 f8 00 16 95 a5 f1 41 30 dc ab 71 80 11  .....A0...q..
0030  02 6f fe 28 00 00 01 01 08 0a 79 09 d1 47 79 09  ..o.(.....y..Gy.
0040  d1 47  ..G
0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 08  .....E.
0010  00 34 e2 34 40 00 40 06 5a 85 7f 00 00 01 7f 00  ..4.4@.@.Z.....
0020  00 01 00 16 a6 f8 30 dc ab 71 95 a5 f1 42 80 10  .....0...q...B..
0030  02 6a fe 28 00 00 01 01 08 0a 79 09 d1 47 79 09  ..j.(.....y..Gy.
0040  d1 47  ..G
0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 08  .....E.
0010  00 34 e2 35 40 00 40 06 5a 84 7f 00 00 01 7f 00  ..4.5@.@.Z.....
0020  00 01 00 16 a6 f8 30 dc ab 71 95 a5 f1 42 80 11  .....0...q...B..
0030  02 6a fe 28 00 00 01 01 08 0a 79 09 d1 4b 79 09  ..j.(.....y..ky.
0040  d1 47  ..G
0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 08  .....E.
0010  00 34 00 00 40 00 40 06 3c ba 7f 00 00 01 7f 00  ..4..@.@.<.....
0020  00 01 a6 f8 00 16 95 a5 f1 42 30 dc ab 72 80 10  .....B0...r..
0030  02 6f d7 5b 00 00 01 01 08 0a 79 09 d1 4b 79 09  ..o.[.....y..ky.
0040  d1 4b  ..K
usuari@server:~$
```

Veiem que els paquets, però el contingut està xifrat, demostrant que sFTP és segur.

La configuració garanteix una transferència segura i amb accés controlat.